

LAB 11

Shell Scripting

Objective

To do programming with Shell Scripts.

Exercise

Write a script which displays Good Morning if hour is less than 12.00, Good Afternoon if hour is less than 5.00p.m, and Good Evening if hour is greater than 5.00p.m.

Script:

```
stewiedev@DESKTOP-RH97L4T: ~/shell_lab11
GNU nano 7.2
#!/bin/bash

hour=$(date +%H)

if [ $hour -lt 12 ]
then
    echo "Good Morning"
elif [ $hour -lt 17 ]
then
    echo "Good Afternoon"
else
    echo "Good Evening"
fi
```

Output:

```
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ nano ex1.sh
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ chmod +x ex1.sh
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ ./ex1.sh
Good Morning
```

Write a script that determines whether number is odd or even.

Script:

```
stewiedev@DESKTOP-RH97L4T: ~/3
GNU nano 7.2
#!/bin/bash

echo -n "Enter Number: "
read num

if [ $(num % 2) -eq 0 ]
then
    echo "Even Number"
else
    echo "Odd Number"
fi
```

Output:

```
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ nano odd_even.sh
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ ./odd_even.sh
Enter Number: 4
Even Number
```

Write a script which takes two numbers as input and asks user choice for multiplication, division, addition, and subtraction. Like (*calculator*), and then calculate the output. Apply suitable checks; if user enters wrong input the program again ask for correct input.

Script:

```
stewiedev@DESKTOP-RH97L4T: ~/shell_lab11
GNU nano 7.2 calc
#!/bin/bash

echo "Enter First Number:"
read a

echo "Enter Second Number:"
read b

echo "1. Addition"
echo "2. Subtraction"
echo "3. Multiplication"
echo "4. Division"

echo -n "Enter Choice: "
read ch

case $ch in
1) echo "Result = $((a+b))" ;;
2) echo "Result = $((a-b))" ;;
3) echo "Result = $((a*b))" ;;
4)
    if [ $b -eq 0 ]
    then
        echo "Division by Zero Not Allowed"
    else
        echo "Result = $((a/b))"
    fi
    ;;
*) echo "Invalid Choice" ;;
esac
```

Output:

```
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ nano calculator.sh
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ chmod +x calculator.sh
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ ./calculator.sh
Enter First Number:
5
Enter Second Number:
7
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter Choice: 3
Result = 35
```

Write a script to print no's as 5, 4, 3, 2, 1 using while loop.

Script:

```
stewiedev@DESKTOP-RH97L4T: ~$ nano
GNU nano 7.2
#!/bin/bash

n=5

while [ $n -ge 1 ]
do
    echo $n
    n=$((n-1))
done
```

Output:

```
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ ./countdown.sh
5
4
3
2
1
```

Write script to print given number in reverse order, for e.g. If no is 123 it must print as 321.

Script:

```
stewiedev@DESKTOP-RH97L4T: ~/sh
GNU nano 7.2
#!/bin/bash

echo -n "Enter Number: "
read num

rev =0

while [ $num -gt 0 ]
do
    digit=$((num % 10))
    rev=$((rev * 10 + digit))
    num=$((num/10))
done

echo "Reverse =$rev"
```

Ouput:

```
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ nano reverse.sh
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ chmod +x reverse.sh
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ ./reverse.sh
Enter Number: 123
Reverse = 321
```

Write script to print given numbers sum of all digits, for e.g. If no is 123 it's sum of all digit will be $1+2+3 = 6$.

Script:

```
stewiedev@DESKTOP-RH97L4T: ~  
GNU nano 7.2  
#!/bin/bash  
  
echo -n "Enter Number: "  
read num  
  
sum=0  
  
while [ $num -gt 0 ]  
do  
    digit=$((num % 10))  
    sum=$((sum + digit))  
    num=$((num / 10))  
done  
  
echo "Sum = $sum"
```

Output:

```
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ ./sumdigits.sh  
Enter Number: 123  
Sum = 6
```

Write Script to find out biggest number from given three no's. No's are supplies as command line argument. Print error if sufficient arguments are not supplied.

Script:

```
stewiedev@DESKTOP-RH97L4T: ~/shell_lab11  
GNU nano 7.2  
#!/bin/bash  
  
if [ $# -ne 3 ]  
then  
    echo "Usage: $0 num1 num2 num3"  
    exit 1  
fi  
  
a=$1  
b=$2  
c=$3  
  
max=$a  
  
if [ $b -gt $max ]  
then  
    max=$b  
fi  
  
if [ $c -gt $max ]  
then  
    max=$c  
fi  
  
echo "Largest Number = $max"
```

Output:

```
stewiedev@DESKTOP-RH97L4T:~/shell_lab11$ ./biggest.sh 10 20 5  
Largest Number = 20
```